UK tree planting targets

Permanent grassland: 5.985m ha × 14% = 837,900 ha
Croppable land: 6.1m ha × 7% = 427,000 ha
Cropped land: 4.1m ha × 7% = 287,000 ha
(croppable and cropped are alternatives)

Observed tree growth

45 Trichobel trees
Diameters converted to dry biomass per tree
Average dry biomass across trees at each visit
Observed ages 1-31

Planting rate and layout

All planted in 2027 OR constant rate between 2027-2050
Cropland: 47.62 trees/ha; 2m strips / 30m rows
Grassland: 166 trees/ha; 2m strips / 10m rows
Alleys retain their previous use

Mean tree biomass over time

Mean tree dry biomass at each measured age
Linear interpolation between measured ages
Use ages 1-30; assume zero biomass at age zero
Same growth assumed across planting layouts

Biomass by planting cohort

For every year in 2027-2056: determine cohort age, area and tree count
Tree count x dry biomass per tree at that age = cohort dry AGB

Aboveground carbon

Dry AGB × 0.4884
Carbon fraction of shoots
(Schulz-Nielsen, et al., 2026)

Belowground carbon

Dry AGB × 0.26 × 0.4690
Root:shoot dry-mass ratio (Cairns et al., 1997) and root carbon fraction
(Schulz-Nielsen, et al., 2026)

Additional soil carbon

Trees on arable:
ΔSOC_{ESM}/ 32 yr = 0.65 Mg C/ha/yr
× strip width × cohort age
Trees on grassland: zero gain
Alleys: zero gain

Sum cohorts and land types

Aggregate AGB C, BGB C and additional SOC across cohorts and land use types
Convert C to CO_2 (× 3.67)

Carbon stocks

- AGB
- AGB + BGB + additional SOC

AGB growth over time

- PAIC: anual AGB carbon gain
- MAIC: average annual gain since planting.
  Calculated for each planting cohort and summed